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Data Science Intern

Week 1

Task : Exploratory Data Analysis

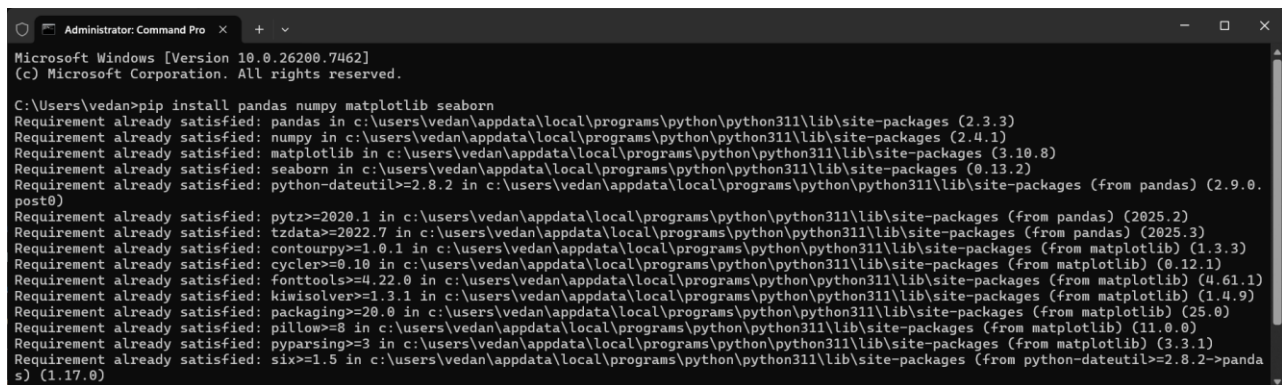
Description: Analyze a real-world dataset (such as Titanic, Iris, or any Kaggle dataset).
Generate insights through visualizations and summary statistics.

Challenges:

- Handle missing and duplicate data
- Visualize key relationships using Matplotlib/Seaborn
- Write a short report summarizing findings

Attached are the sequential screenshots and execution

- Installing the required libraries



```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.26280.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Users\vedan>pip install pandas numpy matplotlib seaborn
Requirement already satisfied: pandas in c:\users\vedan\appdata\local\programs\python\python311\lib\site-packages (2.3.3)
Requirement already satisfied: numpy in c:\users\vedan\appdata\local\programs\python\python311\lib\site-packages (2.4.1)
Requirement already satisfied: matplotlib in c:\users\vedan\appdata\local\programs\python\python311\lib\site-packages (3.10.8)
Requirement already satisfied: seaborn in c:\users\vedan\appdata\local\programs\python\python311\lib\site-packages (0.13.2)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\vedan\appdata\local\programs\python\python311\lib\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\vedan\appdata\local\programs\python\python311\lib\site-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in c:\users\vedan\appdata\local\programs\python\python311\lib\site-packages (from pandas) (2025.3)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\vedan\appdata\local\programs\python\python311\lib\site-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in c:\users\vedan\appdata\local\programs\python\python311\lib\site-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\vedan\appdata\local\programs\python\python311\lib\site-packages (from matplotlib) (4.61.1)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\vedan\appdata\local\programs\python\python311\lib\site-packages (from matplotlib) (1.4.9)
Requirement already satisfied: packaging>=20.0 in c:\users\vedan\appdata\local\programs\python\python311\lib\site-packages (from matplotlib) (25.0)
Requirement already satisfied: pillow>=8 in c:\users\vedan\appdata\local\programs\python\python311\lib\site-packages (from matplotlib) (11.0.0)
Requirement already satisfied: pyparsing>=3 in c:\users\vedan\appdata\local\programs\python\python311\lib\site-packages (from matplotlib) (3.3.1)
Requirement already satisfied: six>=1.5 in c:\users\vedan\appdata\local\programs\python\python311\lib\site-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
```

(All the required libraries were already installed)

- Creating a new Jupyter notebook

Cell wise Jupyter Notebook Code : (EDA.ipynb)

(The code snippets include a comment at the beginning showing the purpose)

```
# Load Libraries
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sea

sea.set(style="whitegrid")
```

```
# Load the dataset
df = sea.load_dataset("titanic")
df.head()
```

```
# Number of rows and columns in dataset
print("Rows:", df.shape[0])
print("Columns:", df.shape[1])
```

```
# Names of the columns present in the dataset
print("Columns in dataset:")
print(df.columns)
```

```
# Detailed information or Physical View of the dataset
df.info()
```

```
# Missing values count
print("Missing values in each column:")
df.isnull().sum()
```

```
# % of missing values
missing_percentage = (df.isnull().sum() / len(df)) * 100
missing_percentage.sort_values(ascending=False)
```

```
# Count of duplicate rows
print("Duplicate rows:", df.duplicated().sum())
```

```
# Basic Statistical Information
df.describe()
```

```
# Handling the missing data
df2 = df.copy()

# Lets replace missing age values with the median
df2["age"] = df2["age"].fillna(df2["age"].median())

# Categorical data, replacing with the mode if not available
df2["embarked"] = df2["embarked"].fillna(df2["embarked"].mode()[0])

# more than 75% data is missing, thus Lets drop the column
df2.drop(columns=["deck"], inplace=True)

df2.isnull().sum()
```

```
# Survival Count (0 -> Not Survived, 1 -> Survived)
df2["survived"].value_counts()
```

```
# Plotting the survival data
sea.countplot(x="survived", data=df2)
plt.title("Survival Count (0 => No, 1 => Yes)")
plt.show()
```

```
# The number of male or female passengers
df2["sex"].value_counts()
```

```
# The count presented on a countplot
sea.countplot(x="sex", data=df2)
plt.title("Gender Count")
plt.show()
```

```
# Seating class of passengers
df2["class"].value_counts()
```

```
# Count of passengers in each class
sea.countplot(x="class", data=df2)
plt.title("Passenger Class Count")
plt.show()
```

```
# Age Distribution Graph
sea.histplot(df2["age"], bins=30, kde=True)
plt.title("Age Distribution")
plt.show()

# Survival count with respect to sex
sea.countplot(x="sex", hue="survived", data=df2)
plt.title("Survival vs Gender")
plt.show()

# The Summary of the analysis
print("Final EDA Summary")
print("-----")

total = len(df2)
survival_rate = df2["survived"].mean() * 100

print("Total passengers:", total)
print(f"Overall Survival Rate: {survival_rate:.2f}%")

print("\nSurvival Rate by Gender:")
print(df2.groupby("sex")["survived"].mean() * 100)

print("\nSurvival Rate by Class:")
print(df2.groupby("class")["survived"].mean() * 100)
```

Output Snippets :

```
Jupyter EDA Last Checkpoint: 8 minutes ago
File Edit View Run Kernel Settings Help
JupyterLab Python 3 (ipykernel)

[26]: # Load Libraries
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sea

sea.set(styles="whitegrid")

[27]: # Load the dataset
df = sea.load_dataset("titanic")
df.head()

[27]:
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	False
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	False
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	yes	True
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	False
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	True

```
[28]: # Number of rows and columns in dataset
print("Rows:", df.shape[0])
print("Columns:", df.shape[1])

Rows: 891
Columns: 15
```

```
Jupyter EDA Last Checkpoint: 10 minutes ago
File Edit View Run Kernel Settings Help
JupyterLab Python 3 (ipykernel)

[29]: # Names of the columns present in the dataset
print("Columns in dataset:")
print(df.columns)

Columns in dataset:
Index(['survived', 'pclass', 'sex', 'age', 'sibsp', 'parch', 'fare',
       'embarked', 'class', 'who', 'adult_male', 'deck', 'embark_town',
       'alive', 'alone'],
      dtype='object')

[30]: # Detailed information or Physical View of the dataset
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
 #   Column        Non-Null Count  Dtype
---  -
 0   survived      891 non-null    int64
 1   pclass        891 non-null    int64
 2   sex           891 non-null    object
 3   age           714 non-null    float64
 4   sibsp         891 non-null    int64
 5   parch         891 non-null    int64
 6   fare          891 non-null    float64
 7   embarked      889 non-null    object
 8   class         891 non-null    category
 9   who           891 non-null    object
10  adult_male    891 non-null    bool
11  deck          203 non-null    category
12  embark_town   889 non-null    object
13  alive         891 non-null    object
14  alone         891 non-null    bool
dtypes: bool(2), category(2), float64(2), int64(4), object(5)
memory usage: 69.7+ KB
```

```
jupyter EDA Last Checkpoint: 11 minutes ago
File Edit View Run Kernel Settings Help Trusted
JupyterLab Python 3 (ipykernel)

[31]: # Missing values count
print("Missing values in each column:")
df.isnull().sum()

Missing values in each column:

[31]: survived      0
      pclass        0
      sex           0
      age          177
      sibsp         0
      parch         0
      fare          0
      embarked      2
      class         0
      who           0
      adult_male    0
      deck         688
      embark_town   2
      alive         0
      alone         0
      dtype: int64
```

```
jupyter EDA Last Checkpoint: 11 minutes ago
File Edit View Run Kernel Settings Help Trusted
JupyterLab Python 3 (ipykernel)

[32]: # % of missing values
missing_percentage = (df.isnull().sum() / len(df)) * 100
missing_percentage.sort_values(ascending=False)

[32]: deck          77.216611
      age          19.865320
      embarked     0.224467
      embark_town  0.224467
      sex          0.000000
      pclass       0.000000
      survived     0.000000
      fare         0.000000
      parch        0.000000
      sibsp        0.000000
      class        0.000000
      adult_male   0.000000
      who          0.000000
      alive        0.000000
      alone        0.000000
      dtype: float64

[33]: # Count of duplicate rows
print("Duplicate rows:", df.duplicated().sum())

Duplicate rows: 107
```

```
jupyter EDA Last Checkpoint: 11 minutes ago
File Edit View Run Kernel Settings Help Trusted
JupyterLab Python 3 (ipykernel)

[34]: # Basic Statistical Information
df.describe()

[34]:
```

	survived	pclass	age	sibsp	parch	fare
count	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

```
[35]: # Handling the missing data
df2 = df.copy()

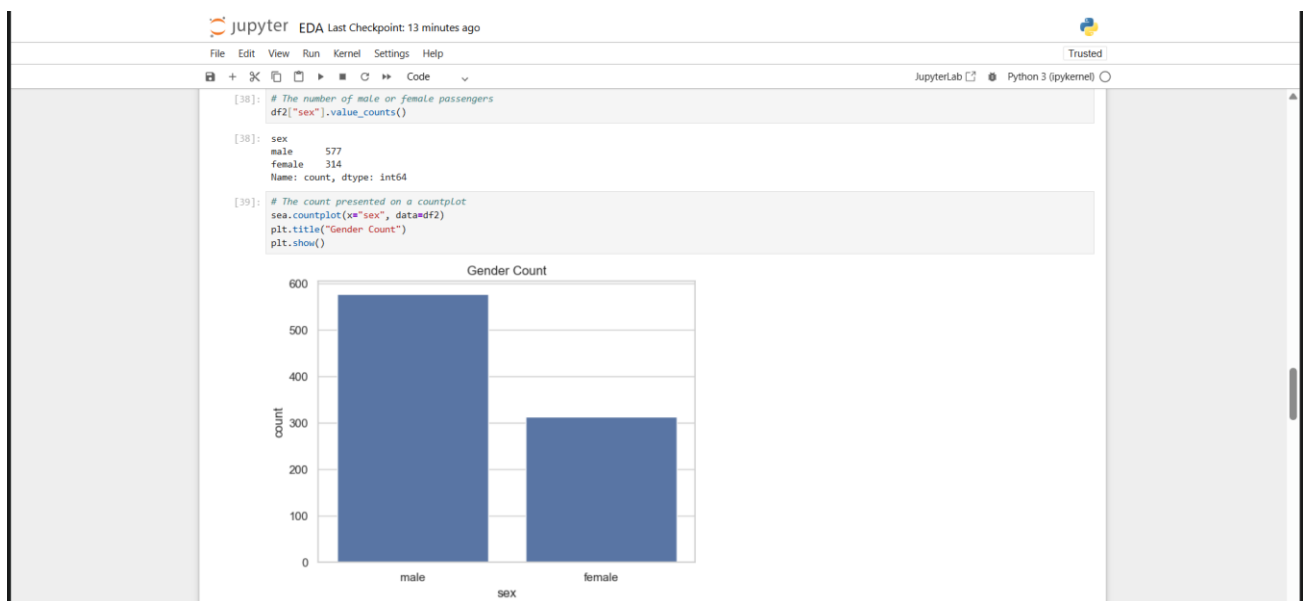
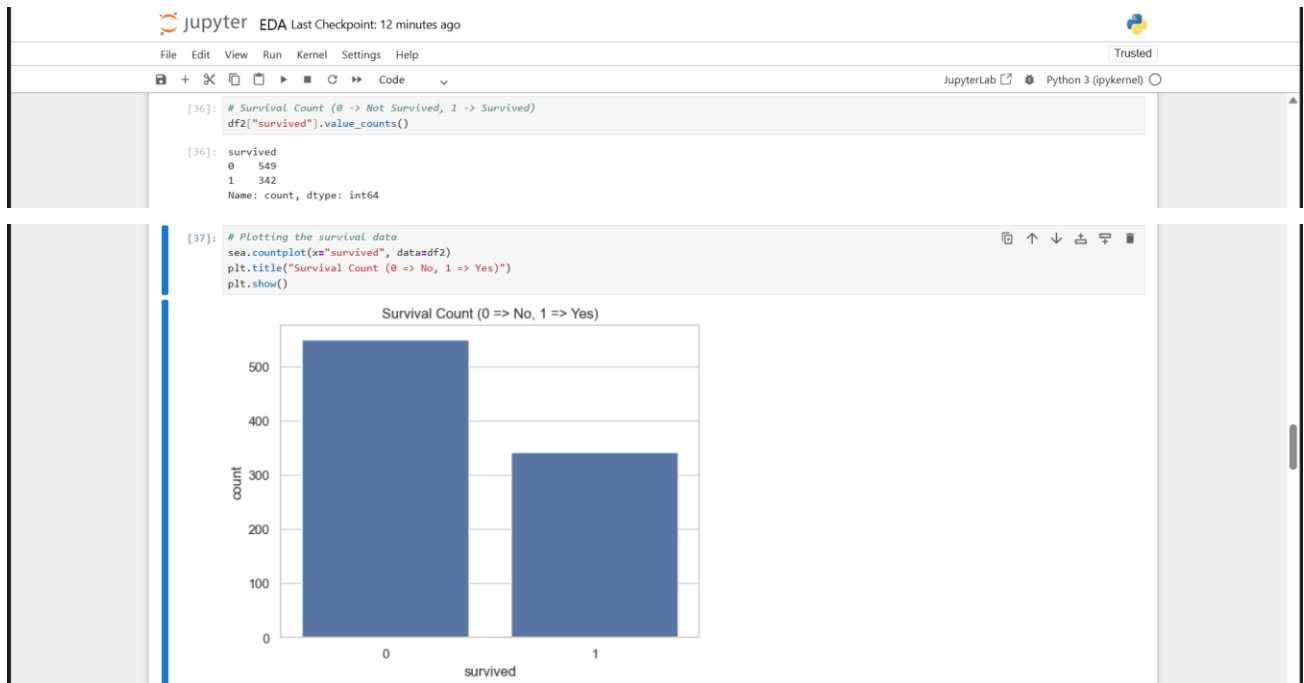
# Lets replace missing age values with the median
df2["age"] = df2["age"].fillna(df2["age"].median())

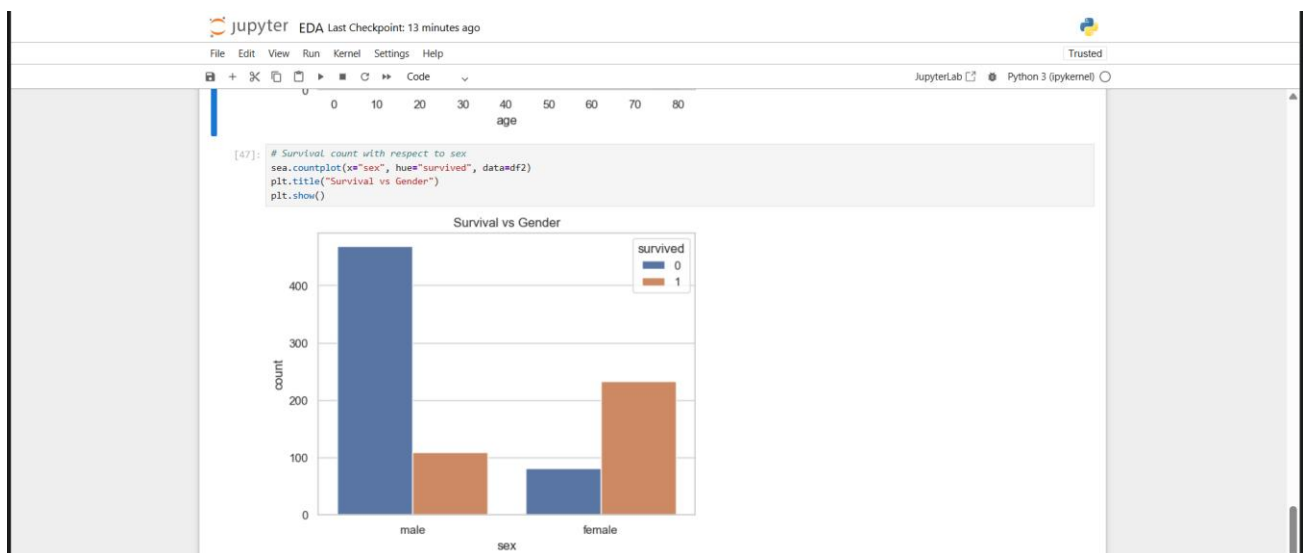
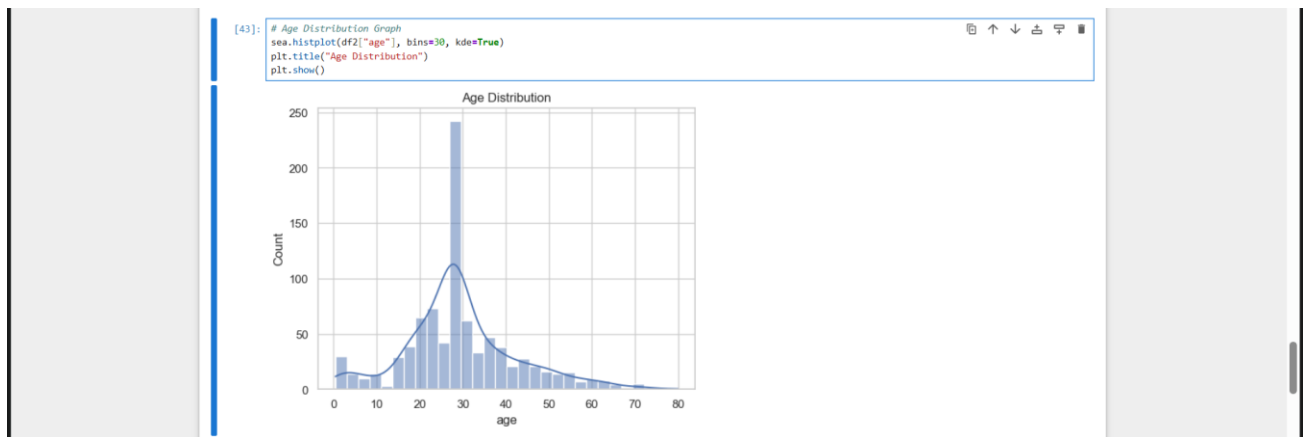
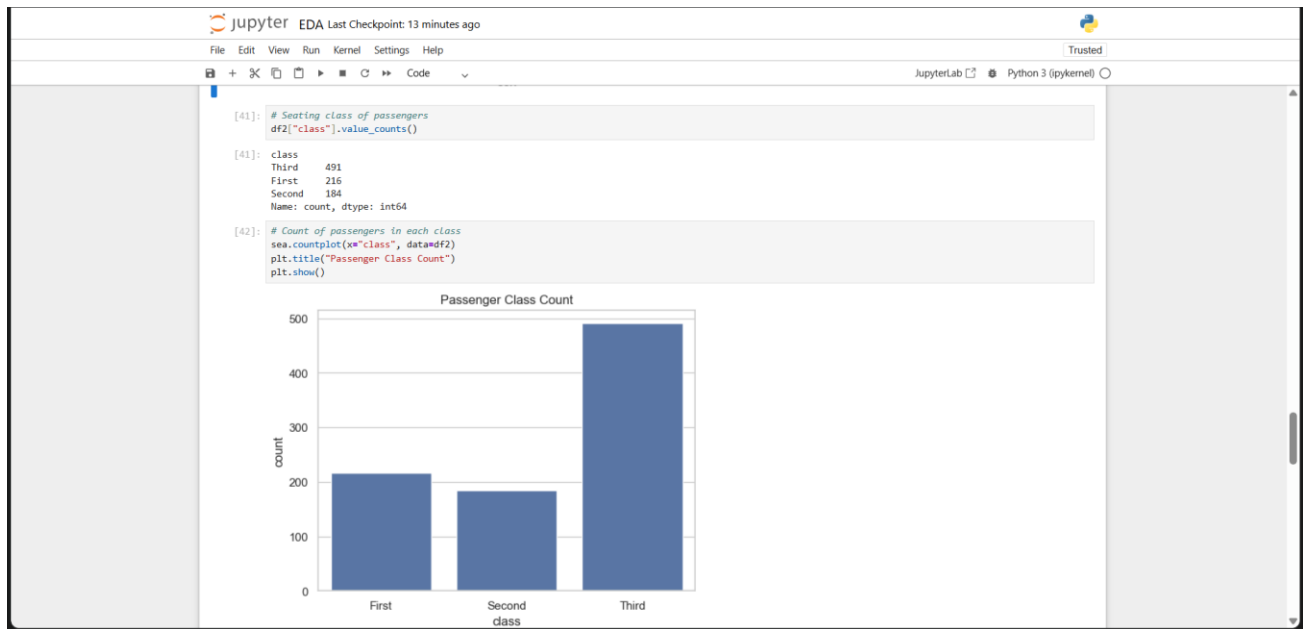
# Categorical data, replacing with the mode if not available
df2["embarked"] = df2["embarked"].fillna(df2["embarked"].mode()[0])

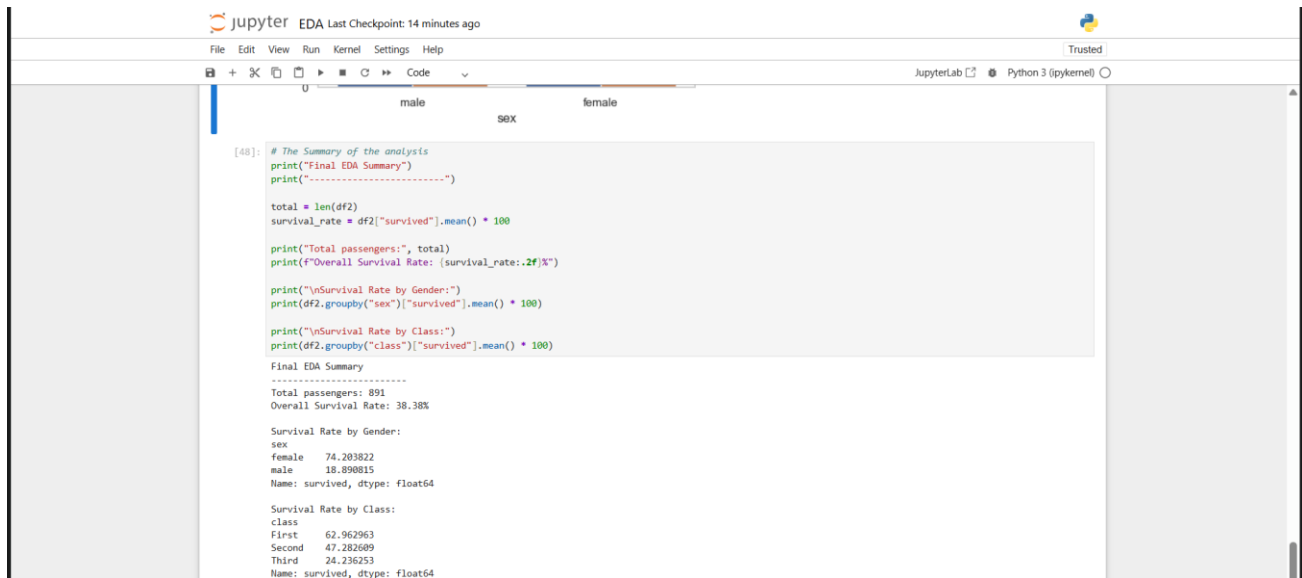
# more than 75% data is missing, thus Lets drop the column
df2.drop(columns=["deck"], inplace=True)

df2.isnull().sum()

[35]: survived      0
      pclass        0
      sex           0
      age          0
      sibsp         0
      parch         0
      fare          0
      embarked      0
      class         0
      who           0
      adult_male    0
      embark_town   2
      alive         0
      alone         0
      dtype: int64
```







The screenshot shows a Jupyter Notebook interface with a code cell containing Python code for data analysis. The code calculates the total number of passengers, the overall survival rate, and survival rates by gender and class. The output displays these statistics in a structured format.

```
[48]: # The Summary of the analysis
print("Final EDA Summary")
print("-----")

total = len(df2)
survival_rate = df2["survived"].mean() * 100

print("Total passengers:", total)
print(f"Overall Survival Rate: {survival_rate:.2f}%")

print("\nSurvival Rate by Gender:")
print(df2.groupby("sex")["survived"].mean() * 100)

print("\nSurvival Rate by Class:")
print(df2.groupby("class")["survived"].mean() * 100)

Final EDA Summary
-----
Total passengers: 891
Overall Survival Rate: 38.38%

Survival Rate by Gender:
sex
female    74.203822
male      18.890815
Name: survived, dtype: float64

Survival Rate by Class:
class
First     62.962963
Second    47.282609
Third     24.235253
Name: survived, dtype: float64
```

The above screenshots show the complete execution of the given challenge on the Titanic dataset by using the techstack : Python, Pandas, Matplotlib, Seaborn and Jupyter Notebook and performing exploratory operations on the data.